

.... (2)

A' B' $A' \cap B'$

Fig.1.31

From (1) and (2), it is verified that $(A \cup B)' = A' \cap B'$

Example 1.26 If $U = \{x : x \in \mathbb{Z}, -2 \leq x \leq 10\}$,

$A = \{x : x = 2p + 1,\ p \in \mathbb{Z}, -1 \leq p \leq 4\}$, $B = \{x : x = 3q + 1,\ q \in \mathbb{Z}, -1 \leq q < 4\}$,

verify De Morgan's laws for complementation.

Solution

Thinking Corner

$A \cap (A \cup B)' =$____

Given $U = \{-2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$,

$A = \{-1, 1, 3, 5, 7, 9\}$ and $B = \{-2, 1, 4, 7, 10\}$

Law (i) $(A \cup B)' = A' \cap B'$

Now, $A \cup B = \{-2, -1, 1, 3, 4, 5, 7, 9, 10\}$

$(A \cup B)' = \{0, 2, 6, 8\}$ ..... (1)

Then, $A' = \{-2, 0, 2, 4, 6, 8, 10\}$ and $B' = \{-1, 0, 2, 3, 5, 6, 8, 9\}$

$A' \cap B' = \{0, 2, 6, 8\}$ ..... (2)

From (1) and (2), it is verified that $(A \cup B)' = A' \cap B'$

Thinking Corner

$(A \cup B)' \cup (A' \cap B) =$____

Law (ii) $(A \cap B)' = A' \cup B'$

Now, $A \cap B = \{1, 7\}$

$(A \cap B)' = \{-2, -1, 0, 2, 3, 4, 5, 6, 8, 9, 10\}$ ..... (3)

Then, $A' \cup B' = \{-2, -1, 0, 2, 3, 4, 5, 6, 8, 9, 10\}$ ..... (4)

From (3) and (4), it is verified that $(A \cap B)' = A' \cup B'$

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